CCE Assignment

Unit-3 Linked list

Student Record Management

1) Problem statement:

Implement a singly linked list in c++ to maintain student records. Each node stores the student's roll number, name and marks. The program should perform the following operations:

- Insert a student record at the begining
- Insert a student record at the end
- Delete a student record
- search for a student Record using roll number.
- Display all student records.

2) objectives:

i) To understand the concept of a singly linked list.

ii) To create and manage student records using dynamic memory.

iii) To perform insertion at the begining and end of the linked list.

iv) To Perform deletion of a student record.

v) To Perform searching using the student's roll number

vi) To display all records stored in the linked list.

vii) To understand dynamic memory allocation using pointers in c++.

3) Algorithm

step 1: start

step 2: Create a structure Node containing:
- Roll number
- student name
- Marks
- Pointer to the next node

step 3: Initialize head = NULL

step 4: Insert at Begining
i) Create a new node
ii) Enter a student details
iii) set newNode->next = head
iv) set head = newNode

step 5: Insert at End
- Create a Node
- Enter student details.
- If the list is empty, mark new node
- otherwise, traverse to the last node
- set the last node's next to the new

step 6: Delete
i) Enter the roll number to delete.
ii) search the linked list for that roll nu
iii) Adjust the links
iv) Delete the selected node.

step 7: search
- Enter the roll number to search.
- Traverse the list from head.

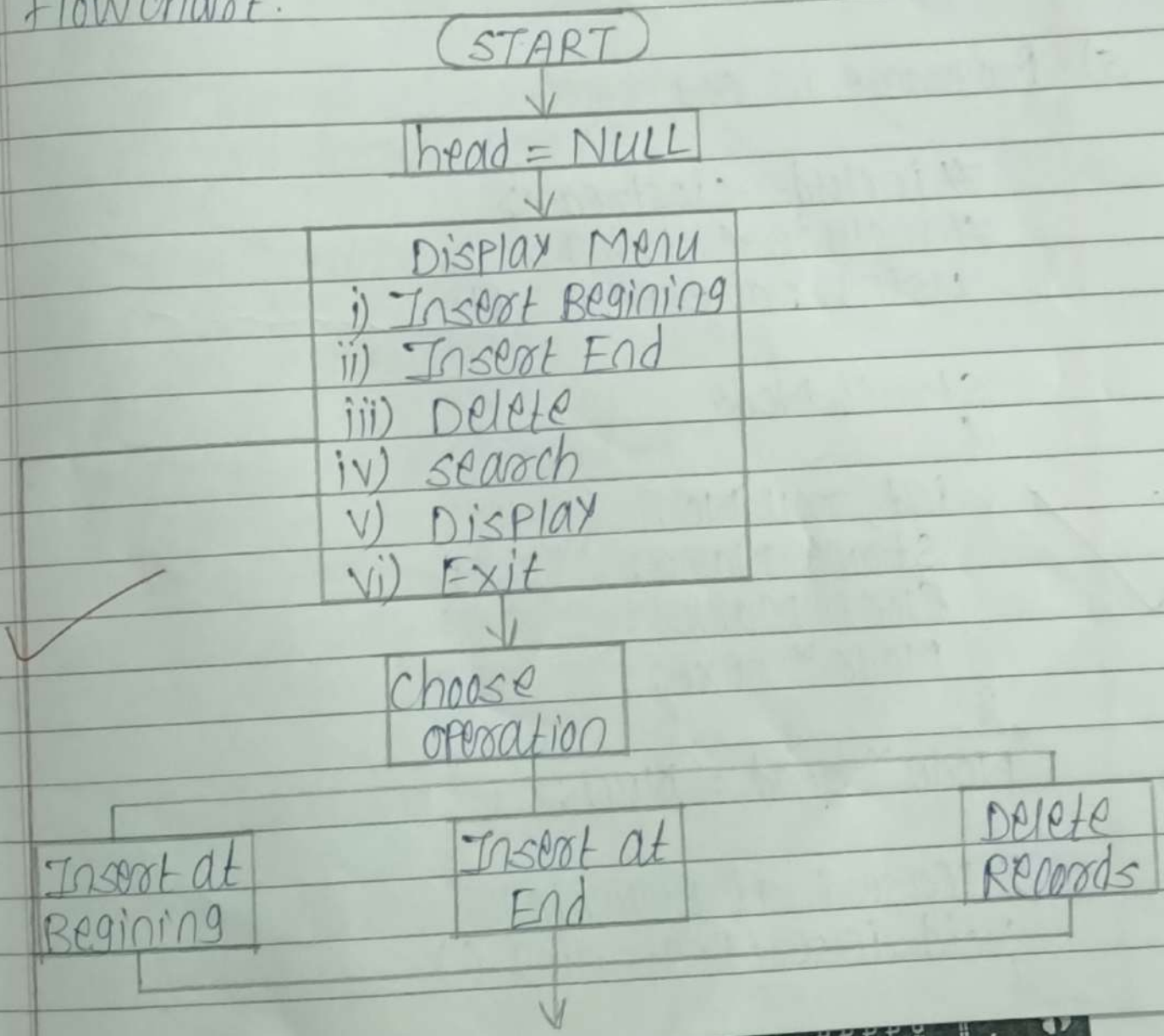
- Compare each node's roll number
- If found, display the student's details.
- Otherwise, display "student not found".

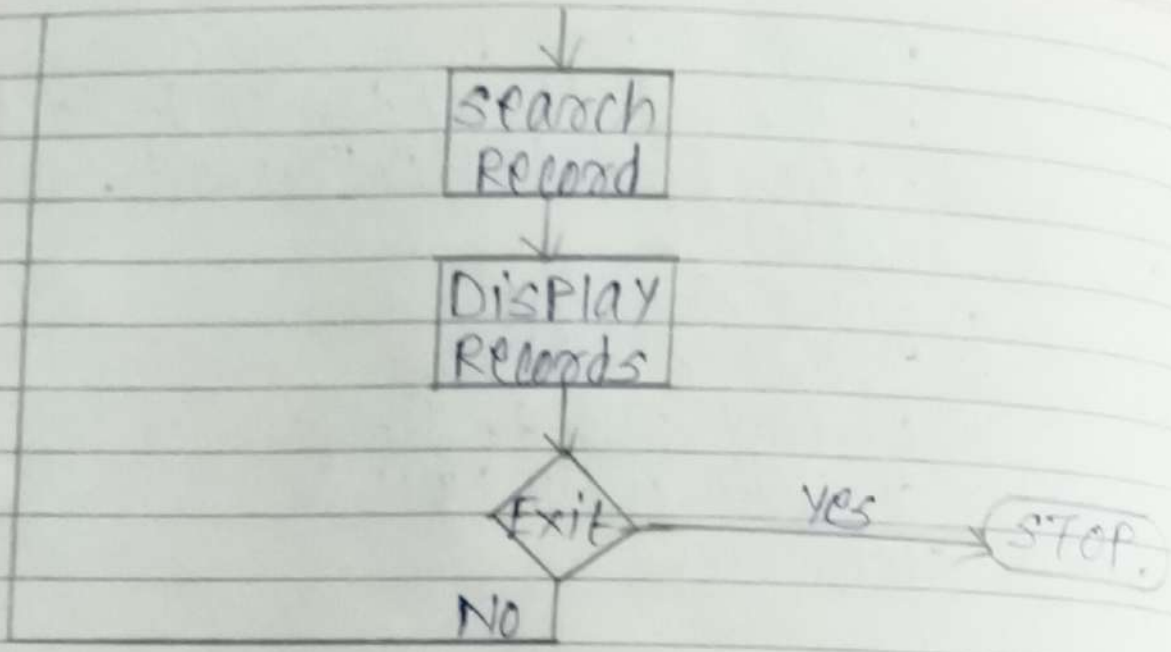
step 8: Display

i) start from head
ii) Traverse each node
iii) Display student details
iv) continue until NULL.

step 9: stop

4) Flowchart:





5) Programe in C++

```
#include <iostream>
#include <string>
 using namespace std;

struct Node
{
  int roll No;
  string Name;
  float Marks
  Node * next;
};
Node * head = NULL;

// Insert at Beginning
 Void insertBeginning ()
{
```

```
    Node* newNode = new Node;
    cout << "Enter Roll No: ";
    cin >> newNode->roll No;
    cout << "Enter Name: ";
    cin >> newNode->Name;
    cout << "Enter Marks: ";
    cin >> newNode->Marks
    newNode->next = head;
    head = new Node;

cout << "Record inserted at beginning.\n";
}
  // Insert at ~~Beginning~~ End
  Void insert End ()
  {
Node* newNode = new Node;
cout << "Enter Roll No: ";
 cin >> new Node -> roll No;

 cout << "Enter Name: ";
 cin >> newNode->name;
 cout << "Enter Marks: ";
 cin >> newNode->marks;
newNode->next = NULL;

if (head == NULL)
{
  head = newNode
}
else
```

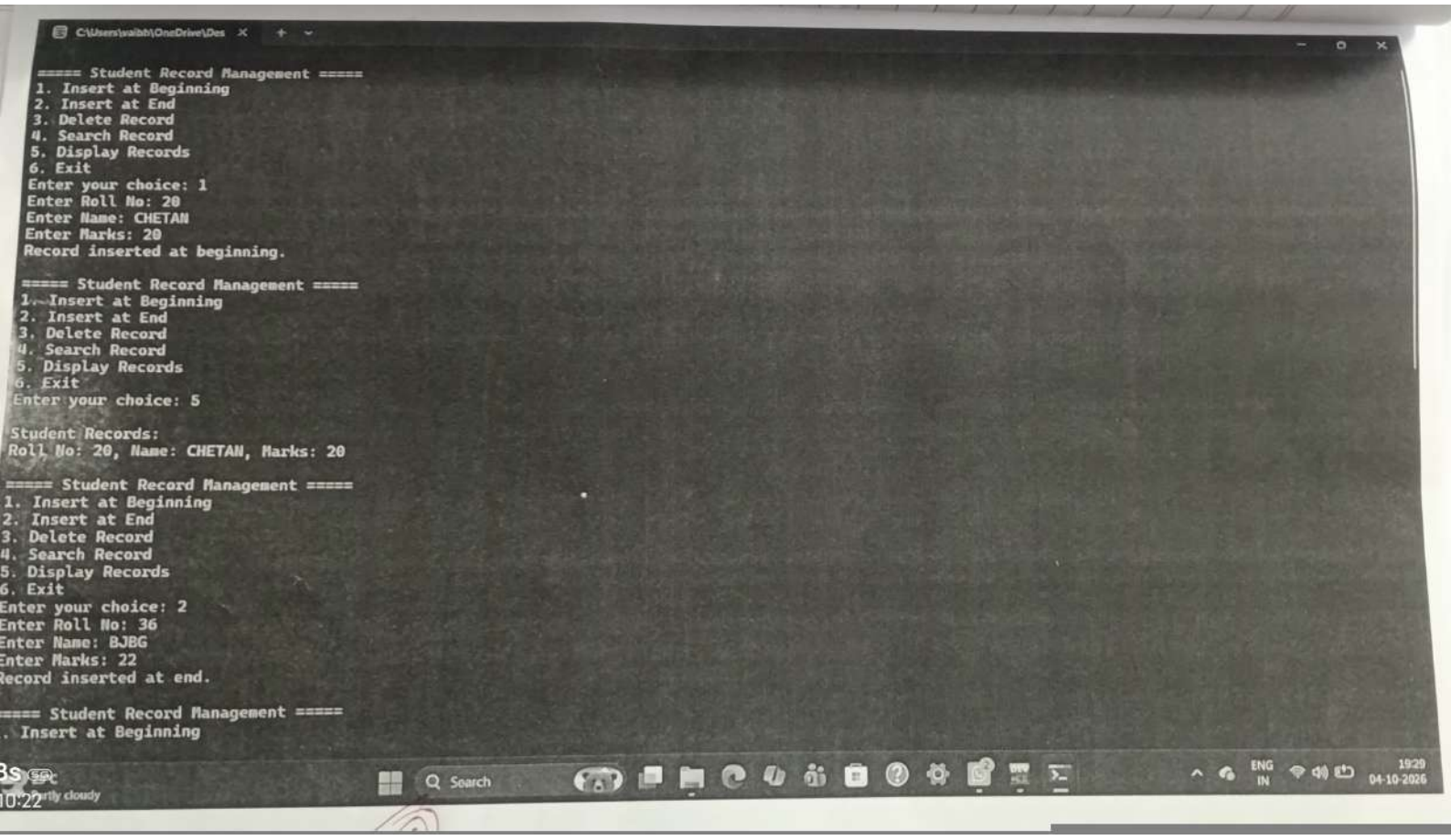
```
{
  Node* temp = head;
  while (temp->next != NULL)
  {
    temp = temp->next;
  }
  temp->next = newNode;
}
cout << " Record inserted at end \n";
}
// Delete Record
void deletRecord()
{
int roll;
cout << " Enter Roll no. to Delete:";
cin >> roll;

Node* temp = head;
Node* PreN = NULL;
while (temp != NULL && temp->roll No != ro
{
PreN = temp;
temp = temp->next;
}
if (temp == NULL)
{
cout << " student Record not found. \n";
return;
}
```

```
if (PreN == Null)
{
head = temp->next;
}
else
{
PreN->next = temp->next;
}
delete temp;
Cout<<" Record deleted successfully:\n";
}
// search Record
Void searchRecord()
{
int roll;
Cout<<" Enter Roll No to search:";
Cin>>roll;
Node* temp = head;
While (temp != NULL)
{
if (temp->roll No == roll)
{
Cout<<"\n student Found!\n";
Cout<<" Roll No: "<< temp->roll No << endl;
Cout<<"Name:" << temp->name << endl;
Cout<<" Marks: "<< temp->marks << endl;
return;
}
temp = temp->next;
}
```

```
cout << " student record not found \n";
}
// Display Records
void display()
{
if (head == NULL)
{
cout << " No students Records: \n";
while (temp != NULL)
{
cout << "Roll No: " << temp -> roll No
cout << "Name: " << temp -> next
cout << " Marks: " << temp -> marks <<
temp = temp -> next;
}
}
int main()
{
int choice;
do
{
cout << "\n ==== Student Record Management ==
cout << "1. Insert at Beginning \n";
cout << "2. Insert at End \n";
cout << "3. Delete Record \n";
cout << "4. search Record \n";
cout << "5. Display Record \n";
cout << "6. Exit \n";
```

```
cout << " Enter Your choice : ";
cin >> choice;
switch (choice)
{
case 1:
        Insert Beginning();
        break;
case 2: insertEnd();
        break;
case 3: delete Record();
        break;
case 4: search Record();
        break;
case 5: display();
        break;
case 6:
        cout << " Program ended \n";
    break;
    default:
cout << " Invalid choice ! \n";
    }
}
while (choice != 6);
return 0;
}
```


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===== Student Record Management =====
1. Insert at Beginning
2. Insert at End
3. Delete Record
4. Search Record
5. Display Records
6. Exit
Enter your choice: 1
Enter Roll No: 20
Enter Name: CHETAN
Enter Marks: 20
Record inserted at beginning.
===== Student Record Management =====
1. Insert at Beginning
2. Insert at End
3. Delete Record
4. Search Record
5. Display Records
6. Exit
Enter your choice: 5
Student Records:
Roll No: 20, Name: CHETAN, Marks: 20
===== Student Record Management =====
1. Insert at Beginning
2. Insert at End
3. Delete Record
4. Search Record
5. Display Records
6. Exit
Enter your choice: 2
Enter Roll No: 36
Enter Name: BJBG
Enter Marks: 22
Record inserted at end.
===== Student Record Management =====
. Insert at Beginning
Search
ENG
IN
19:29
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